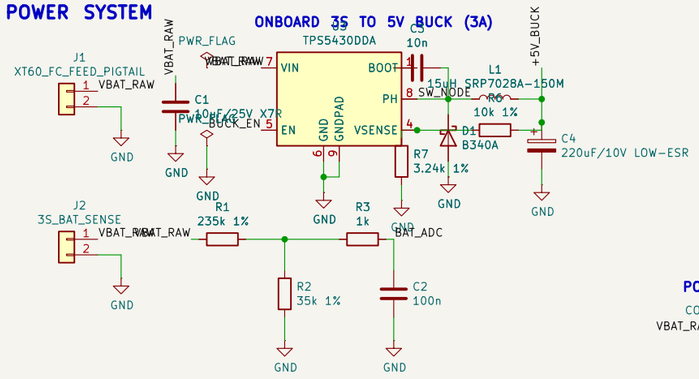
ESP32 FLIGHTSTACK — INTEGRATED 4-LAYER FLIGHT CONTROLLER

Architecture: onboard ESP32-WROOM-32E, MPU-9250, 3S-to-5V buck, 3.3V LDO, GPS, iBUS and four PWM ESC outputs.

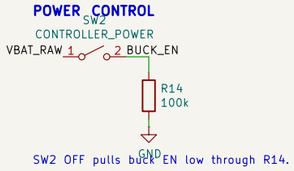
SAFETY: Raw 3S LiPo input only. First power-up must use a current-limited bench supply; test motors without propellers.

POWER SYSTEM

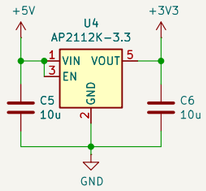
ONBOARD 3S TO 5V BUCK (3A)



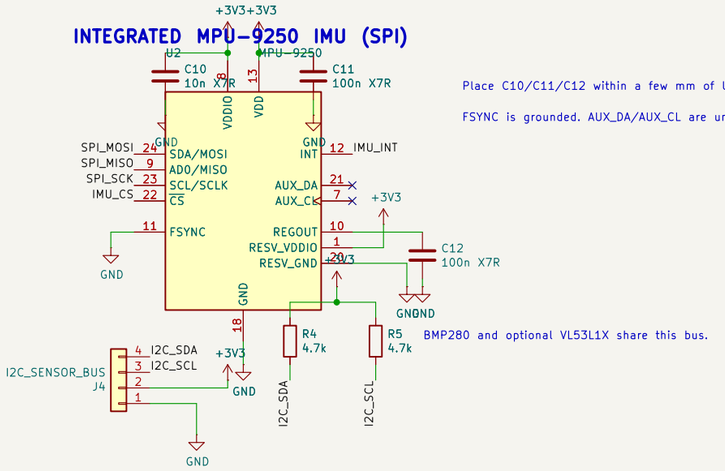
POWER CONTROL



SW2 OFF pulls buck EN low through R14.



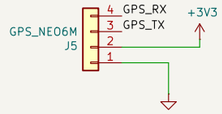
INTEGRATED MPU-9250 IMU (SPI)



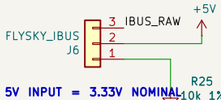
Place C10/C11/C12 within a few mm of U2. Keep U2 away from high-current motor and BEC paths.

FSYNC is grounded. AUX_DA/AUX_CL are unused. Pin 1 follows datasheet: reserved, tied to VDDIO.

BMP280 and optional VL53L1X share this bus.



GPS TXD -> GPS_RX (GPIO23); GPS RXD <- GPS_TX (GPIO17)

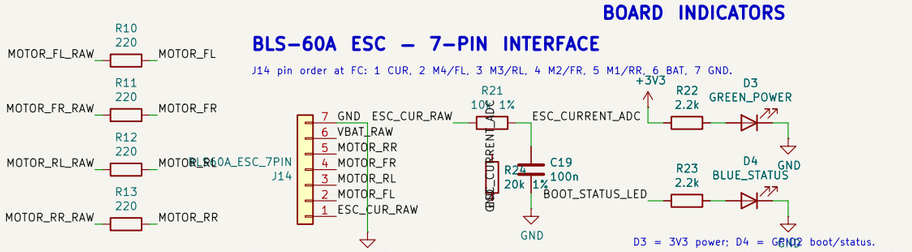


Confirm receiver signal is 3.3 V logic before connection.

FOUR PWM ESC OUTPUTS — SIGNAL AND GROUND ONLY

BOARD INDICATORS

BLS-60A ESC — 7-PIN INTERFACE



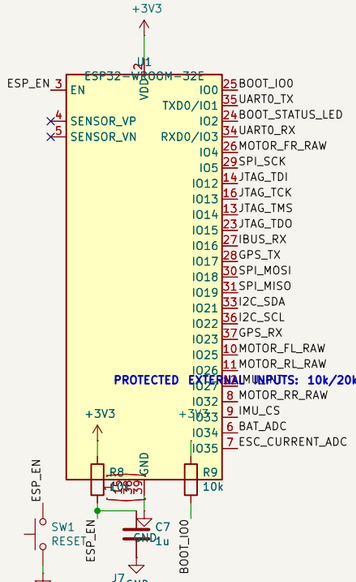
J14 pin order at FC: 1 CUR, 2 M4/FL, 3 M3/RL, 4 M2/FR, 5 M1/RR, 6 BAT, 7 GND.

BAT is a low-current FC supply/sense feed. XT60 and motor current terminate on the ESC, never through this PCB.

D3 = 3V3 power; D4 = GPIO2 boot/status.

Motor map: FL GPIO25 (CCW), FR GPIO4 (CW), RL GPIO26 (CW), RR GPIO32 (CCW).

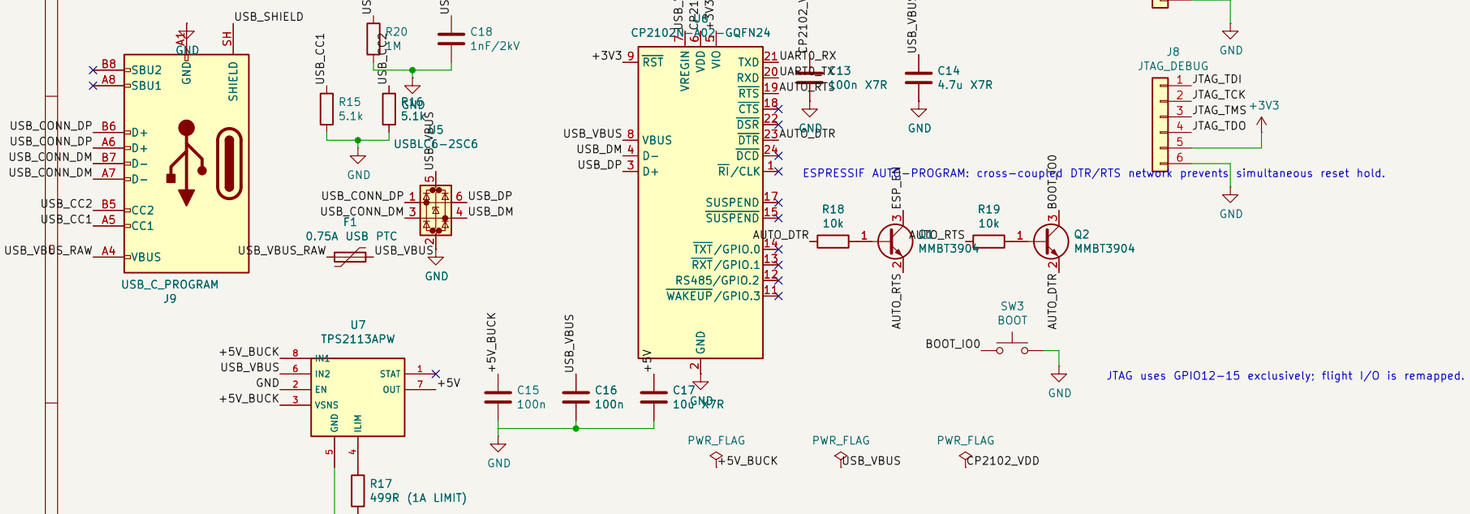
ESC PORTS: signal + ground only. Do not connect ESC BEC outputs.



PROTECTED EXTERNAL INPUTS: 10k/20k DIVIDERS — 5V INPUT = 3.33V NOMINAL

USB-C PROGRAMMING AND PROTECTED POWER MUX

USB VBUS and battery-derived 5 V are isolated by U7; simultaneous connection is permitted.



JTAG uses GPIO12–15 exclusively; flight I/O is remapped.

U7 IN1 = battery buck, IN2 = protected USB. Reverse/cross conduction are blocked internally.

ASSEMBLY / BRING-UP NOTES

1. Inspect assembly, then apply raw 3S input from a current-limited bench supply.
2. Confirm +3V3, then verify the battery ADC node with a multimeter.
3. Confirm MPU-9250 WHO_AM_I and BMP280 detection before connecting ESCs.
4. First motor tests must be performed without propellers.
5. Document the PCB/IMU axis orientation in firmware before flight testing.

Firmware source: https://github.com/durveshpathak1989/ESP32-FlightStack

Prototype flight controller — verify on bench before flight
BLS-60A 7-pin ESC, protected USB/battery mux, onboard power
Pin mapping matches RC_FlightController.ino
ESP32-WROOM-32E + MPU-9250 + BMP280 + GPS + iBUS + 4 PWM ESC
durveshpathak1989 / ESP32-FlightStack

Sheet: /
File: ESP32_FlightController.kicad_sch

Title: ESP32 FlightStack Integrated Flight Controller

Size: A3	Date: 2026-07-20	**Rev: 1.2**
KiCad E.D.A. 10.0.3		Id: 1/1